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Unit – 2 ⇝ Vector Space (Tutorial Manual)

Method 1 ⇝ Vector Space

Examples of Method-1: Vector Spaces

A	1	Prove that the set $V = \mathbb{R}^n$ with the standard operations of addition and scalar multiplication is a vector space.
A	2	Prove that the set $V = \mathbb{P}_n$ with the standard operations of addition and scalar multiplication is a vector space.
B	3	Prove that the set $V = \mathbb{M}_{m\times n}$ with the standard operations of addition and scalar multiplication is a vector space.
C	4	Show that the set $V = \mathbb{R}^3$ with the standard vector addition and scalar multiplication define as $c \cdot (u_1, u_2, u_3) = (0, 0, cu_3)$ is not a vector space. **Hint: No, 10th condition is not satisfied**
C	5	Show that the set of all pairs of real numbers of the form $(1, x)$ with the operations defined as $(1, x_1) + (1, x_2) = (1, x_1 + x_2)$ and $k(1, x) = (1, kx)$ is a vector space.
C	6	Show that the set of all pairs of real numbers of the form $(1, x)$ with the operations $(1, y) + (1, y') = (1, y + y')$ and $k(1, y) = (1, ky)$ is a vector space.
B	7	Check whether $V = \mathbb{R}^2$ is a vector a space with respect to the operations $(u_1, u_2) + (v_1, v_2) = (u_1 + v_1 - 2, u_2 + v_2 - 3)$ and $\alpha(u_1, u_2) = (\alpha u_1 + 2\alpha - 2, \alpha u_2 - 3\alpha + 3), \alpha \in \mathbb{R}$. **Answer: No, because 7th and 8th conditions are not satisfied**
B	8	Check that $V = \{ (x, y, z) / x, y, z \in \mathbb{R} \} = \mathbb{R}^3$,with operations $(x_1, y_1, z_1) * (x_2, y_2, z_2) = (x_2, y_1 + y_2, z_1)$ and $k \odot (x_1, y_1, z_1) = (kx_1, ky_1, kz_1)$ is a vector space or not? **Answer: No, 2nd, 4th, 5th & 8th conditions are not satisfied**

B	9	Show that $V = \left\{ \begin{bmatrix} a & b \\ c & 1 \end{bmatrix} / a, b, c \in R \right\}$ is vector space under the operations: $\begin{bmatrix} a & b \\ c & 1 \end{bmatrix} + \begin{bmatrix} d & e \\ f & 1 \end{bmatrix} = \begin{bmatrix} a+d & b+e \\ c+f & 1 \end{bmatrix}$ and $k \cdot \begin{bmatrix} a & b \\ c & 1 \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & 1 \end{bmatrix}$.
A	10	Check $V = \left\{ \begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} / a, b \in R \right\}$ is a vector space or not under standard matrix addition & scalar multiplication. **Answer: No, 1st, 4th, 5th and 6th conditions are not satisfied**
B	11	Prove that all upper triangular matrices of order 2 over $\mathbb{R}$ is a vector space with standard matrix addition and scalar multiplication.
A	12	Check whether the set $\mathbb{P}_3 = \{a_0 + a_1x + a_2x^2 + a_3x^3 / a_0, a_1, a_2, a_3 \in \mathbb{R}\}$ is a vector space or not with standard operations. **Answer: Yes, Vector Space**

Method 2 ⇝ Subspaces

Examples of Method-2: Subspaces

A	1	Show that $W = \{ (x, y) / \ x = 2y \}$ is a subspace of the vector space $\mathbb{R}^2$.
B	2	Check whether $W = \{(x, y) / xy \geq 0\}$ is a subspace of $\mathbb{R}^2$ or not? **Answer: No, 1st condition is not satisfied**
B	3	Is $W = \{(x, y) / x = 2y + 3\}$ a subspace of $\mathbb{R}^2$? **Answer: No, both conditions are not satisfied**
A	4	Check whether $W = \{(0, 0, z) / z \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$ or not? **Answer: Yes, Subspace**
C	5	Does $W = \{(x, y, z) / x^2 + y^2 + z^2 = 1\}$ is a subspace of $\mathbb{R}^3$ with the standard operations? **Answer: No, 1st & 2nd conditions are not satisfied**
B	6	Check whether $W = \{(x, y, z) / x^2 + y^2 + z^2 \leq 1\}$ is a subspace of $\mathbb{R}^3$ or not? **Answer: No, 1st & 2nd conditions are not satisfied**
A	7	Prove that the set $W = \{(a, 2a, a + 1)/a \in \mathbb{R}\}$is not vector subspace of $\mathbb{R}^3$. **Answer: No, 1st & 2nd conditions are not satisfied**
B	8	Determine $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} / b = a, c = d = -a \right\}$ is a subspace of $\mathbb{M}_2$ or not. **Answer: Yes, Subspace**
B	9	Prove that $W = \left\{ \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} / a + c = 0, b + d + f = 0 \right\}$ is a subspace of $\mathbb{M}_{2\times3}$.
A	10	Determine $W = \left\{ \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} / a = 2c + 1 \right\}$ is a subspace of $\mathbb{M}_{2\times3}$ or not. **Answer: No, 1st & 2nd conditions are not satisfied**

A	11	Prove that $W = \{a_1x + a_2x^2 + a_3x^3 / a_1, a_2, a_3 \in \mathbb{R}\}$ is a subspace of $\mathbb{P}_3$.
B	12	Prove that $W = \{a_0 + a_1x + a_2x^2 \ / \ a_0 = 2\}$ is not a subspace of $\mathbb{P}_2$? **Answer: No, 1st & 2nd conditions are not satisfied**
B	13	Determine whether $W = \{a_0 + a_1x + a_2x^2 \ / \ a_0 = a_1 + a_2\}$ is a subspace of $\mathbb{P}_2$ or not? **Answer: Yes, Subspace**
B	14	Prove that $W = \{a_0 + a_1x + a_2x^2 + a_3x^3 \ / \ a_0 + a_1 + a_2 + a_3 = 0 \ \& \ a_i \in \mathbb{R}\}$ is a subspace of $\mathbb{P}_3$.

Method 3 ⇝ Linear Combination and Span

Examples of Method-3: Linear Combination and Span

A	1	Express $a = (5, -1, 9)$ as a linear combination of $v_1 = (2, 9, 0)$, $v_2 = (3, 3, 4)$ & $v_3 = (1, 2, 1)$. **Answer:** $\mathbf{a = -v_1 + 2v_2 + v_3}$
A	2	Express a vector $u = (7, 4, -3)$ as a linear combination of $v = (1, -2, -5)$ and $w = (2, 5, 6)$. **Answer:** $\mathbf{u = 3v + 2w}$
A	3	Consider the vectors $u = (1, 2, -1)$ and $v = (6, 4, 2)$ in $\mathbb{R}^3$. Show that $w = (9, 2, 7)$ is a linear combination of u and v. **Answer:** $\mathbf{w = -3u + 2v}$
B	4	Determine whether $v = \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix}$ can be generated by a linear combination of $u_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$, $u_2 = \begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix}$ and $u_3 = \begin{bmatrix} 1 \\ -5 \\ -7 \end{bmatrix}$. **Answer:** $\mathbf{v = \frac{37}{28} u_1 + \frac{11}{28} u_2 + \frac{-3}{28} u_3}$
B	5	Show that the matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is a linear combination of $X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $Y = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $Z = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. **Answer:** $\mathbf{A = (-1)X + (-1)Y + (2)Z}$
B	6	The matrix $\begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ & $\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$. **Answer:** $\mathbf{A = 2A_1 + A_2 + A_3}$

C	7	Check matrix A is linear combination of A_1, A_2 and A_3 or not, where $A = \begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$, $A_1 = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $A_3 = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$. **Answer: Not linear combination**
B	8	Express $p(x) = 6 + 11x + 6x^2$ as a linear combination of the following: $p_1(x) = 2 + x + 4x^2$, $p_2(x) = 1 - x + 3x^2$ and $p_3(x) = 3 + 2x + 5x^2$. **Answer: $p(x) = 4p_1(x) - 5p_2(x) + 1p_3(x)$**
A	9	Determine whether the given vectors $v_1 = (1, -1, 1)$, $v_2 = (0, 1, 2)$ and $v_3 = (3, 0, -1)$span $\mathbb{R}^3$ or not. **Answer: Yes**
A	10	Define Span and Determine whether the given vectors $v_1 = (1, 1, 2)$, $v_2 = (1, 0, 1)$ and $v_3 = (2, 1, 3)$ span $\mathbb{R}^3$ or not. **Answer: No**
A	11	Does $S = \{(2, -1, 3), (4, 1, 2), (8, -1, 8)\}$ span $\mathbb{R}^3$? Justify with reason. **Answer: No**
A	12	Determine whether the given vectors $v_1 = (3, 1, 4)$, $v_2 = (2, -3, 5)$, $v_3 = (5, -2, 9)$ and $v_4 = (1,4, -1)$ span $\mathbb{R}^3$ or not. **Answer: No**
B	13	Prove that following matrices are span $\mathbb{M}_{22}$. Where, $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $A_3 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ and $A_4 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
C	14	Are $p_1(x) = 1 - x + 2x^2$, $p_2(x) = 5 - x + 4x^2$ and $p_3(x) = -2 - 2x + 2x^2$ span $\mathbb{P}_2$? **Answer: No**

Method 4 ⇝ Linear Dependence and Independence

Examples of Method-4: Linear Dependence and Independence

A	1	Is S = {(0, 0, 0)} linearly dependent? Why? **Answer: Yes, zero vector**
A	2	Is S = {(1, 2, 3)} linearly independent? Why? **Answer: Yes, non − zero singleton set**
A	3	Is S = {(2, 5, −7), (1, 2, 3)} is linearly dependent or independent. Why? **Answer: Linearly independent because $\bar{u} \neq k\bar{v}$**
A	4	Is the following set linearly dependent or not? S = {(−2, 0, 1), (3, 2, 5), (6, −1, 1), (7, 0, −2)} **Answer: Linearly dependent because $\dim(\mathbb{R}^3) < \lvert S \rvert$**
A	5	Prove that (2, 1, 2), (1, 2, 2) and (2, 2, 1) are linearly independent.
A	6	Let $v_1 = (1, -1, 0)$, $v_2 = (0, 1, -1)$, $v_3 = (0, 0, 1)$ be elements in $\mathbb{R}^3$. Verify the set of vectors v_1, v_2, v_3 is linearly independent or linearly dependent. **Answer: Linearly independent**
B	7	Check dependence of (1, 1, 2, 1),(1, 0, 0, 2), (4, 6, 8, 6), (0, 3, 2, 1). **Answer: Linearly dependent**
B	8	Check dependence of (0, 0, 2, 2), (3, 3,0, 0), and (1, 1, 0, −1). **Answer: Linearly independent**
C	9	Determine vectors (1, 1, 1, 3), (1, 2, 3, 4) and (2, 3, 4, 7) are linearly independent or not. If they are linearly dependent then find relation between them. **Answer: Linearly dependent, $\bar{u}_3 = \bar{u}_1 + \bar{u}_2$**
B	10	Prove that $p_1(x) = 1 - 2x + x^2$, $p_2(x) = 4 - x + 3x^2$ are not linearly dependent.

C	11	Show that $S = \{1 - t - t^3, -2 + 3t + t^2 + 2t^3, 1 + t^2 + 5t^3\}$ is linearly independent in $\mathbb{P}_3$.
B	12	Check whether the set of following matrices is linearly dependent? $\left\{\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}, \begin{bmatrix}2 & 3\\1 & 2\end{bmatrix}, \begin{bmatrix}3 & 1\\2 & 1\end{bmatrix}, \begin{bmatrix}2 & 2\\1 & 1\end{bmatrix}\right\}$. **Answer: Linearly independent**
B	13	Check $\begin{bmatrix}1 & 1\\1 & 2\end{bmatrix}, \begin{bmatrix}1 & 0\\0 & 2\end{bmatrix}, \begin{bmatrix}0 & 3\\1 & 2\end{bmatrix}$ and $\begin{bmatrix}2 & 6\\4 & 6\end{bmatrix}$ are linearly dependent or independent in $\mathbb{M}_2$. **Answer: Linearly dependent**
A	14	Check $A = \begin{bmatrix}-3 & 4\\2 & 0\end{bmatrix}$ and $B = \begin{bmatrix}3 & -4\\-2 & 0\end{bmatrix}$ are linearly dependent or independent in $\mathbb{M}_2$. **Answer: Linearly dependent**

Method 5 ⇝ Basis and Dimension

Examples of Method-5: Basis and Dimension

A	1	Prove that $S = \{(1,2,1),(2,9,0),(3,3,9)\}$ forms a basis for $\mathbb{R}^3$.
A	2	Determine whether the following set of vectors forms a basis for $\mathbb{R}^3$ or not. $\{(1,1,2),(1,2,5),(5,3,4)\}$. **Answer: No**
A	3	Determine whether the following set of vectors forms a basis for $\mathbb{R}^4$ or not. $\{(1,-1,8),(1,-2,0),(0,-9,7)\}$. **Answer: No**
B	4	Show that $S = \left\{\begin{bmatrix}1 & 2\\ 1 & -2\end{bmatrix}, \begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}, \begin{bmatrix}0 & 2\\ 3 & 1\end{bmatrix}, \begin{bmatrix}0 & 0\\ -1 & 2\end{bmatrix}\right\}$ is a basis for $\mathbb{M}_2$.
B	5	Show that the following set of vectors is a basis for $\mathbb{M}_2$. $S = \left\{\begin{bmatrix}3 & 6\\ 3 & -6\end{bmatrix}, \begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}, \begin{bmatrix}0 & -8\\ -12 & -4\end{bmatrix}, \begin{bmatrix}1 & 0\\ -1 & 2\end{bmatrix}\right\}$.
B	6	Let $p_1 = 1 + x$, $p_2 = 1 + x^2$ & $p_3 = x + x^2$. Show that set $S = \{p_1, p_2, p_3\}$ is a basis for $\mathbb{P}_2$.
C	7	Show that the following set is basis for $\mathbb{P}_3$. $\{1 + 4x - 2x^2, 2x + x^2, -3 + x + x^2, 5 - 2x - 3x^3\}$
B	8	Check whether $S = \{-4 + x + 3x^2, 6 + 5x + 2x^2, 8 + 4x + x^2\}$ forms a basis for $\mathbb{P}_2$ or not? **Answer: Yes**
B	9	Let $p_1 = 1 - 3x + 2x^2$, $p_2 = 1 - x + 4x^2$, $p_3 = 1 - 7x$. Prove $\{p_1, p_2, p_3\}$ is a basis for $\mathbb{P}_2$.
A	10	Write a standard basis and find dimension of the following vector spaces. (1). $\mathbb{R}^3$, (2). $\mathbb{P}_4$, and (3). $\mathbb{M}_{3\times4}$.

B	11	Write a standard basis and find dimension of the following vector spaces. (1). $\mathbb{R}^n$, (2). $\mathbb{P}_n$, and (3). $\mathbb{M}_{m\times n}$.
B	12	Let $\mathbb{R}^2$ be a vector space and S be the subspace given by the set of vectors $(a, 0)$. Find a basis for S and evaluate dim(S). **Answer: Basis $= \{(1, 0)\}$, Dimension $= 1$**
B	13	Let $u = (0, 0, 3), v = (-5, 0, 0)$ and $w = (0, 0, 9)$ span a subspace S of $\mathbb{R}^3$. Determine the dimension of S. **Answer: Basis $= \{(0, 0, 3), (-5, 0, 0)\}$, Dimension $= 2$**
C	14	Let $\mathbb{P}_3$ be a vector space and S be the subspace spanned by the set of vectors $\{1 - x^2,\ 1 + x\}$. Find a basis for S and evaluate dim(S). **Answer: Basis $= \{1 - x^2, 1 + x\}$, Dimension $= 2$**

Method 6 ⇝ Row Space, Column Space, and Null Space

Examples of Method-6: Row Space, Column Space, and Null Space

A	1	Find the basis and dimension of row space of $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 9 & -1 \\ -3 & 8 & 3 \\ -2 & 3 & 2 \end{bmatrix}$. **Answer:** $\{(1 \quad 2 \quad -1), (1 \quad 9 \quad -1)\}, 2$
B	2	Define column space. Find basis for column space of $A = \begin{bmatrix} 2 & -3 & 6 \\ 1 & -1 & 5 \\ -1 & 2 & 0 \\ 4 & 1 & 1 \end{bmatrix}$. **Answer:** $\left\{\begin{bmatrix} 2 \\ 1 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 5 \\ 0 \\ 1 \end{bmatrix}\right\}, 3$
A	3	Find basis and dimension for row and column space of $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 0 \end{bmatrix}$. **Answer:** $\{(1 \quad 2 \quad 3), (0 \quad 1 \quad -3)\}, 2$ $\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right\}, 2$
C	4	Find a basis for the row and column space of $A = \begin{bmatrix} 1 & 4 & 5 & 4 \\ 2 & 9 & 8 & 2 \\ 2 & 9 & 9 & 7 \\ -1 & -4 & -5 & -4 \end{bmatrix}$. **Answer:** $\left\{\begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 9 \\ 9 \\ -4 \end{bmatrix}, \begin{bmatrix} 5 \\ 8 \\ 9 \\ -5 \end{bmatrix}\right\}, 3$
B	5	Find the basis for null space of $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$. **Answer:** $\left\{\begin{bmatrix} 16 \\ 9 \\ 1 \end{bmatrix}\right\}$

C	6	Find the basis and dimension for null space of $\begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$. **Answer:** $\left\{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}\right\}, 2$
B	7	Determine the basis, dimension for the solution space of the system $x - 3y + z = 0, 2x - 6y + 2z = 0, 3x - 9y + 3z = 0$. **Answer: Basis** $= \{(3, \ 1, \ 0), (-1, \ 0 \ \ 1)\}$, **dimension** $= 2$
A	8	Determine basis & the dimension for the solution space of the system $m + 2n - 3p = 0, 2m + 5n + p = 0, m - n + 2p = 0$. **Answer: No basis, dimension** $= 0$
B	9	Determine the dimension and basis for the solution space of the system $x_1 + 2x_2 + x_3 + 3x_4 = 0, 2x_1 + 5x_2 + 2x_3 + x_4 = 0$ and $x_1 + 3x_2 + x_3 - x_4 = 0$. **Answer:** $\{(-1, \ 0, 1, \ 0)\}, \ 1$
C	10	Determine a basis and dimension of the solution space of the homogeneous system $2x_1 + 2x_2 - x_3 + x_5 = 0, -x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0, x_1 + x_2 - 2x_3 - x_5 = 0$ & $x_3 + x_4 + x_5 = 0$. **Answer: Basis** $= \{(-1, 0, -1, 0, 1), (-1, 1, 0, 0, 0)\}, 2$

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